

Education

Harvard University

Anticipated PhD in Statistics, advised by Prof. Marinka Zitnik

Cambridge, MA

Aug. 2023–Present

University of Oxford

MSc in Statistics, advised by Prof. Yee Whye Teh

Oxford, UK

Oct. 2021–June 2023

Stanford University

B.S. in Computer Science and B.A. in Sociology with Honors – GPA: 3.9

Stanford, CA

Sept. 2017–June 2021

Honors & Awards

2023	James Mill Peirce Fellowship , Harvard University	Cambridge, MA
2023	Graduate Research Fellowship , National Science Foundation	Cambridge, MA
2021	Rhodes Scholarship , Rhodes Trust	Oxford, UK
2021	Outstanding Senior Thesis Award and Firestone Medal , Stanford University	Stanford, CA
2020	Spotlight Talk , Computational Biology Workshop, ICML	Vienna, AT
2017	G.R.E.A.T. Award , NHGRI, National Institutes of Health	Bethesda, MD

Publications

Published and accepted

Count Bridges enable Modeling and Deconvolving Transcriptomic Data. Nic Fishman, Gokul Gowri, Tanush Kumar, Jiaqi Lu, Valentin De Bortoli, Jonathan S. Gootenberg, and Omar Abudayyeh. *International Conference on Learning Representations*, 2026.

Rethinking Repeatability in Observational Social Science. Jonathan Ben-Menachem, Ari Galper, and Nic Fishman. *Sociological Methods & Research (accepted)*, 2026.

Generative Distribution Embeddings. Nic Fishman, Gokul Gowri, Peng Yin, Jonathan Gootenberg, and Omar Abudayyeh. *Advances in Neural Information Processing Systems*, 2025.

Partisan disparities in the funding of science in the United States. Alexander C Furnas, Nic Fishman, Leah Rosenstiel, and Dashun Wang. *Science*, 2025.

The fragility of fairness: Causal sensitivity analysis for fair machine learning. Jake Fawkes, Nic Fishman, Mel Andrews, and Zachary C Lipton. *Advances in Neural Information Processing Systems*, 2024.

Estimating Vote Choice in U.S. Elections with Approximate Poisson-Binomial Logistic Regression. Nic Fishman and Evan Rosenman. *NeurIPS 2024 Workshop on Optimization for Machine Learning*, 2024.

Human mobility networks reveal increased segregation in large cities. Hamed Nilforoshan, Wenli Looi, Emma Pierson, Blanca Villanueva, Nic Fishman, Yiling Chen, John Sholar, Beth Redbird, David Grusky, and Jure Leskovec. *Nature*, 2023.

Diffusion models for constrained domains. Nic Fishman, Leo Klarner, Valentin De Bortoli, Emile Mathieu, and Michael Hutchinson. *Transactions on Machine Learning Research*, 2023.

Metropolis sampling for constrained diffusion models. Nic Fishman, Leo Klarner, Emile Mathieu, Michael Hutchinson, and Valentin De Bortoli. *Advances in Neural Information Processing Systems*, 2023.

Change we can believe in: Structural and content dynamics within belief networks. Nic Fishman and Nicholas T Davis. *American Journal of Political Science*, 2022.

Should attention be all we need? The epistemic and ethical implications of unification in machine learning. Nic Fishman and Leif Hancox-Li. *Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency*, 2022.

Systematic characterization of generative models for de novo design of regulatory DNA. Nic Fishman, Avanti Shrikumar, Georgi Marinov, and Anshul Kundaje. *ICML Computational Biology Workshop (Spotlight)*, 2020.

Preprints and working papers

Editorial Screening when Science is Cheap. Nic Fishman and Gabriel Sekeres. *Working paper*, 2026.

Distribution-Conditioned Transport. Nic Fishman, Gokul Gowri, Paolo LB Fischer, Marinka Zitnik, Omar Abudayyeh, and Jonathan Gootenberg. *arXiv:2603.04736*, 2026.

Making (Global) Criminal Procedure: Empire, the End of Justice, and the Rise of Efficiency. Nic Fishman. *Stanford University*, 2021.

Teaching Experience

Harvard University, STAT 221: Computational Statistics: Monte Carlo and Optimization Section Leader; overall section-leader evaluation: 5.00/5 (8 responses)	Cambridge, MA Spring 2026
Harvard University, STAT 236: Sparse Inference, and Network and Text Analysis Section Leader; overall section-leader evaluation: 4.88/5 (8 responses)	Cambridge, MA Fall 2024

Presentations

2025	Generative modeling on the space of empirical measures , Boston Diffusion Day	Cambridge, MA
2025	Generative modeling on the space of empirical measures , Harvard Statistics Seminar	Cambridge, MA
2025	False Promises and False Premises of Fair Machine Learning , Columbia Algorithms and Society Seminar	New York, NY
2025	Predicting perturbations at single-cell resolution , Cell Circuits and Epigenomics Seminar	Cambridge, MA
2023	Constrained Riemannian Diffusion Modeling , Thesis Defense	Oxford, UK
2022	Invited talk, Should attention be all we need? , Stanford Human Centered Artificial Intelligence	Stanford, CA
2022	Invited talk, Should attention be all we need? , DeepMind Sociotechnical Systems	London, England
2022	Making (Global) Criminal Procedure , Law and Society Association Annual Meeting	Lisbon, Portugal
2021	Structural and content dynamics within belief networks , American Political Science Association Annual Meeting	Seattle, WA

Research Experience

Harvard Medical School, Harvard University PhD research with Prof. Marinka Zitnik	Cambridge, MA Apr. 2024–Present
- Building agentic systems to automate statistical method development, with applications in biology and health-care. - Developing multiscale models for biological systems spanning genomics, single-cell RNA-seq, and spatial transcriptomics, with applications to perturbation prediction. - Developing EHR-based methods for treatment-effect estimation and policy optimization at health-system scale, including ongoing work with Epic on clinically interpretable decision support.	
Harvard Medical School, Harvard University Research with Profs. Jonathan Gootenberg and Omar Abudayyeh	Cambridge, MA May 2024–Present
- Developing transport and deconvolution methods for single-cell perturbation modeling, enabling generalization across unseen perturbations and cellular contexts. - Designing transformer architectures for set-valued representations of cell populations to support fine-grained <i>in silico</i> experimentation and response prediction. - Led benchmarking and validation efforts for out-of-distribution prediction in single-cell sequencing, unifying existing methods and evaluation metrics.	

Institute for Quantitative Social Science, Harvard University

Research with Prof. Kosuke Imai

Cambridge, MA

June 2020–Sept. 2024

- Constructed a non-parametric framework for causal inference under general interference using regression.
- Developed efficient algorithms and theory for non-parametric monotone regression for estimation and inference in the general interference framework, with applications in large-scale network and spatial experiments.
- Worked on adaptive methods for pure exploration in combinatorial designs in generalized linear models.

Department of Statistics, University of Oxford

Research in OxCSML Group under Yee Whye Teh

Oxford, UK

Sept. 2021–June 2023

- Built constrained diffusion and flow-matching models for cyclic peptides and antibody loops, connecting generative modeling to molecular design problems.
- Developed techniques for extending diffusion models to constrained domains in general Riemannian geometries.

Center for Science of Science & Innovation, Northwestern University

Research Fellow

Evanston, IL

June 2021–Sept. 2023

- Synthesized huge international datasets on scientific grantmaking and publication to generate a picture of the landscape and political implications of scientific funding.
- Leveraged LLMs to develop fine-grained topic classifications across grants and paper abstracts to understand when grant-making institutions lead scientific investigation as opposed to following what is already being studied.

Economics Department, Massachusetts Institute of Technology

Research Assistant for Prof. Victor Chernozhukov

Cambridge, MA

June 2021–June 2022

- Developed methods for dynamic treatment effect estimation and policy learning in healthcare.
- Specifically extended g-estimation to non-linear blip functions using tools from minimax optimization.

Stanford Center on Poverty and Inequality and Stanford Network Analysis Project

Research Intern for Prof. Jure Leskovec and Prof. David Grusky

Stanford, CA

Jan. 2020–June 2021

- Used path crossings to study socioeconomic stratification in the United States, developing insights about heterogeneity in segregation across different social contexts and across different income deciles.
- Worked on algorithms to identify when individuals cross paths in time and space using massive GPS data.

Kundaje Lab, Stanford AI Lab

Research Intern for Prof. Anshul Kundaje

Stanford, CA

Jan. 2018–June 2021

- Created modular system for generating DNA sequences using GANs/VAEs/Transformers and optimizing these models to produce sequences with specified properties (level of gene expression or chromatin accessibility).
- Developed a k-NN algorithm to evaluate (1) the fidelity of samples from generative models and (2) the robustness of neural network predictions on regression outputs (extending existing work on classification).

Work Experience

Knowledge Graph Team, Google

Research Intern

San Francisco, CA

Summer 2022

- Developed LLM and knowledge-graph methods for search, including KG-informed representations and author-expertise models for multimedia topic modeling.

Data for Progress

Data Scientist

New York, NY

June 2018–May 2021

- Led polling infrastructure and survey analysis, including database, reporting, and weighting systems used in state and national polling.
- Developed NLP and ecological-inference methods for policy research, including topic models for open-ended responses and models of demographic bias in news.